Task 1 – Lab Document Practice & Prompting Report

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Course

AI Prompting in 2026 – Lab Practice

Introduction

The purpose of this lab was to understand how modern AI systems respond to different prompting techniques and how prompt quality affects output quality. During the lab, I practiced all major concepts including context engineering, retrieval modes, reasoning prompts, multimodal inputs, brainstorming workflows, data analysis prompting, model comparison, and prompt iteration.

The lab demonstrated that AI performance depends heavily on the quality of instructions, context, constraints, and feedback provided by the user.

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1. What I Learned From the Lab

A. Context Improves Results Dramatically

One of the biggest lessons was that AI performs much better when detailed context is provided.

Example

Weak Prompt:

"Which phone should I buy?"

Strong Prompt:

"Help me choose a phone. My budget is $300, I mainly use WhatsApp, take family photos, and need good battery life."

Observation

The first response was generic.

The second response was personalized and much more useful.

Learning

AI behaves like a knowledgeable assistant, but it can only use the information provided to it.

B. Confidence Does Not Mean Accuracy

The lab showed that AI can sometimes sound confident even when information is uncertain or outdated.

Example

Questions involving:

- Current news
- Legal requirements
- Country-specific regulations
- Recent events

required verification through web search.

Learning

I learned to:

- Ask for sources
- Request citations
- Instruct AI to admit uncertainty

instead of accepting answers blindly.

C. Different Retrieval Modes Produce Different Results

The lab introduced three retrieval modes:

1. Pretrained Knowledge

Used for stable information such as:

- History
- Literature
- Science concepts

2. Web Search

Used for:

- Latest news
- Weather
- Recent developments

3. Deep Research

Used for:

- Detailed reports
- Evidence-based comparisons
- Multi-source analysis

Learning

The wording of the prompt determines whether the AI answers from memory or performs research.

D. Structured Prompts Produce Better Outputs

Using structured prompts with:

- Context
- Constraints
- Desired output format

improved answer quality significantly.

Example Template

I need help with [task].

Here is what you need to know:

- Fact 1
- Fact 2
- Fact 3

What I want back:

- Desired output

Learning

Simple structure is more effective than complicated wording

E. Reasoning Prompts Improve Decision Making

Adding instructions such as:

"Think hard before answering"

resulted in:

- Better trade-off analysis
- More detailed reasoning
- Improved recommendations

Learning

Reasoning prompts are useful for complex decisions but unnecessary for simple factual questions.

F. Prompt Iteration Is Extremely Important

The Brainstorm–Iterate Loop was one of the most useful techniques.

Process:

1. Generate options
2. Provide feedback
3. Generate improved options
4. Refine the best result

Learning

The first answer is rarely the best answer.

The best results come from multiple iterations.

G. Multimodal AI Extends Beyond Text

The lab demonstrated that AI can analyze:

- Images
- Handwritten notes
- Receipts
- Whiteboards

Learning

AI can save significant time extracting information from images, but human verification is still required for accuracy.

H. AI Can Build Small Applications

Using Goal–Input–Output prompts, AI successfully generated:

- Calculators
- Timers
- Simple utilities

Learning

Clear requirements are more important than programming knowledge when creating simple AI-generated applications.

I. Data Analysis Requires Code Verification

One important lesson was that AI may estimate numerical results unless explicitly instructed to run code.

Learning

Always use prompts such as:

"Write and run code. Show me the code used."

This reduces calculation errors and increases reliability.

J. Different Models Have Different Strengths

Comparing ChatGPT, Claude, and Gemini showed differences in:

- Writing style
- Creativity
- Reasoning
- Structure

Learning

There is no universally best AI model.

Different tasks may benefit from different models.

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2. How My Prompting Improved During the Process

At the beginning of the lab, my prompts were short and vague.

Example:

"Write an email."

The output was generic and required significant editing.

After practicing the techniques, I started including:

- Audience
- Goal
- Tone
- Constraints
- Desired format

Example:

"Write a short professional email to my manager requesting a meeting reschedule. Keep the tone polite and friendly. Limit to 100 words."

The quality improved significantly.

My prompts became:

- More specific
- More structured
- More goal-oriented

which produced more accurate results.

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3. Prompting Techniques That Helped Me the Most

1. Context Engineering

Most impactful technique.

Adding relevant information consistently improved outputs.

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2. Brainstorm–Iterate Loop

Instead of accepting the first response, I learned to:

- Request alternatives
- Provide feedback
- Refine results

This generated much better outcomes.

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3. Think Hard Prompting

Useful for:

- Decision making
- Trade-off analysis
- Strategic recommendations

The responses were deeper and more thoughtful.

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4. Explicit Output Formatting

Examples:

- Bullet points
- Tables
- Checklists
- Step-by-step plans

This made outputs easier to use immediately.

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5. Asking for Sources and Citations

Helpful for:

- Research

- News
- Professional work

This increased trust and accuracy.

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4. Challenges Faced and How I Solved Them

Challenge 1: Generic Responses

Problem

Short prompts produced broad answers.

Solution

Added context, constraints, and audience information.

Result

More relevant and personalized outputs.

Challenge 2: Overly Confident Answers

Problem

AI sometimes sounded certain about information that could be outdated.

Solution

Asked for:

- Sources
- Citations
- Uncertainty disclosure

Result

More trustworthy responses.

Challenge 3: Weak First Drafts

Problem

Initial outputs were often average.

Solution

Used the Brainstorm–Iterate Loop.

Result

Significant improvement after multiple revisions.

Challenge 4: Data Analysis Accuracy

Problem

AI occasionally estimated calculations.

Solution

Required code execution and code visibility.

Result

More reliable numerical outputs.

Challenge 5: Model Differences

Problem

Different models generated different results.

Solution

Compared outputs across multiple AI systems.

Result

Learned which model performed best for specific tasks.

Conclusion

This lab demonstrated that prompting is a practical skill rather than a collection of tricks. The quality of AI output depends largely on the quality of instructions provided by the user.

The most valuable lessons were:

1. Context is critical.
2. Prompt iteration improves results dramatically.
3. AI confidence should not be confused with accuracy.
4. Different tasks require different prompting strategies.
5. Verification through sources, citations, and code execution is essential.

Overall, the lab significantly improved my ability to communicate with AI systems and produce higher-quality, more reliable outputs.